
From BM25 to Corrective RAG: Evaluating Retrieval Strategies for Heterogeneous Text-and-Table Documents

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Abstract

1 A RAG pipeline is only as good as the documents it retrieves, yet retrieval on
2 corpora that mix prose and tables has received little empirical study. MTEB
3 and BEIR evaluate on homogeneous text, where dense retrieval generally wins;
4 whether that ordering survives on documents whose answers live in table cells is
5 unclear. We evaluate nine retrieval configurations on T²-RAGBench, a financial
6 QA benchmark of 23,088 queries over 7,318 SEC-filing documents, together with
7 a BGE-M3 robustness panel covering encoder choice and a ColBERTv2 follow-up
8 driven by our error analysis. Reported metrics include Recall@*k*, MRR, nDCG,
9 end-to-end Number Match, and paired bootstrap significance tests.

10 Four results stand out. BM25 beats text-embedding-3-large on nearly every
11 metric, inverting the MTEB/BEIR ordering. Hybrid RRF followed by a cross-
12 encoder reranker reaches Recall@5 = 0.816 and MRR@3 = 0.605, well ahead of
13 any single-stage method. HyDE and multi-query expansion hurt rather than help:
14 hallucinated numbers pull the query embedding away from documents with the
15 exact figure. Of the 31% of queries whose gold document is missed at rank 5,
16 73% fail because the answer sits in a table cell that the encoder cannot recover
17 from linearized text. We release the evaluation code, prompts, FAISS indices, and
18 per-query outputs.

1 Introduction

20 In a RAG system [Lewis et al., 2020], a retriever supplies the context an LLM conditions on. If
21 the retriever misses the relevant document, no amount of prompt engineering recovers the answer:
22 retrieval sets a hard upper bound on the whole pipeline. Most RAG research nevertheless treats
23 retrieval as a solved substrate and focuses attention on prompting, decoding, and judging.

24 **The gap for documents combining text and tables.** That assumption breaks down on documents
25 that mix prose and tables. Financial filings, earnings reports, and regulatory documents are built this
26 way: the prose carries context, the tables carry the numbers. Answering “*What was the 2019 year-*
27 *over-year revenue growth?*” requires finding both the right document and the right cell inside it. The
28 benchmarks most practitioners consult, MTEB [Muennighoff et al., 2023] and BEIR [Thakur et al.,
29 2021], evaluate on homogeneous text, where dense retrieval usually outperforms sparse baselines.
30 Whether that ordering survives a shift to heterogeneous documents is an open empirical question,
31 and our main finding is that it does not.

32 We use T²-RAGBench [Strich et al., 2026], a financial QA benchmark of 23,088 queries over 7,318
33 documents. The original paper evaluated six retrieval methods with two metrics; we evaluate nine
34 retrieval configurations together with a BGE-M3 robustness panel that varies encoder choice and

35 a ColBERTv2 follow-up driven by our error analysis, and report the full retrieval and generation
36 metric suite along with paired bootstrap significance tests.

37 **Contributions.**

- 38 1. A head-to-head evaluation of nine retrieval configurations on T²-RAGBench, cover-
39 ing sparse, dense, hybrid, cross-encoder-reranked, query-side, index-side, and adaptive
40 pipelines, with paired bootstrap significance tests, together with a BGE-M3 robustness
41 panel that varies encoder choice and a ColBERTv2 follow-up evaluating late-interaction
42 retrieval against the error taxonomy.
- 43 2. Evidence that rankings on MTEB and BEIR do not transfer to documents combining text
44 and tables: BM25 beats a strong dense encoder (text-embedding-3-large) on nearly
45 every metric.
- 46 3. An error analysis of 100 sampled retrieval failures showing that 73% share a single cause:
47 the answer is a table cell that serialized-text encoders cannot locate. We argue this is the
48 next concrete open problem in the area.
- 49 4. Two negative results: HyDE and multi-query expansion degrade retrieval on numerical
50 financial queries, and CRAG-style correction does not close the gap to hybrid fusion.
- 51 5. Ablations on fusion type and reranker candidate depth, plus a released evaluation frame-
52 work with prompts, indices, and per-query outputs.

53 **2 Related Work**

54 **RAG retrieval methods.** BM25 remains competitive in zero-shot settings [Thakur et al., 2021].
55 Dense dual-encoders [Karpukhin et al., 2020] and their successors (E5 [Wang et al., 2022], E5-
56 Mistral [Wang et al., 2024], BGE-M3 [Chen et al., 2024a]) advance MTEB rankings through con-
57 trastive pretraining. Hybrid retrieval fuses sparse and dense signals via RRF [Cormack et al., 2009],
58 consistently improving recall by 15–30% [Li et al., 2025]. Two-stage reranking with cross-encoders
59 further refines results [Gao et al., 2024]. Query-side methods include HyDE [Gao et al., 2023]
60 and RAG-Fusion [Raudaschl, 2024]. Index-side methods include Contextual Retrieval [Anthropic,
61 2024] and HyPE [Vake et al., 2025]. Adaptive approaches like CRAG [Yan et al., 2024] trigger
62 corrective searches when confidence is low.

63 **QA over documents that mix prose and tables.** HybridQA [Chen et al., 2020] introduced multi-
64 hop reasoning over linked tables and passages. In the financial domain, FinQA [Chen et al., 2021],
65 TAT-QA [Zhu et al., 2021], and ConvFinQA [Chen et al., 2022] target numerical reasoning over
66 hybrid contexts. A recent survey [Nan et al., 2025] finds that retrieval of heterogeneous context
67 remains the primary bottleneck.

68 **RAG benchmarks.** BEIR [Thakur et al., 2021] established zero-shot retrieval evaluation across 18
69 datasets. Recent RAG benchmarks include RGB [Chen et al., 2024b], CRAG [Yang et al., 2024], and
70 RAGBench [Friel et al., 2024]. Li et al. [2025] study RAG design choices at COLING 2025. None
71 focus on retrieval method comparison over documents that mix prose and tables. T²-RAGBench
72 [Strich et al., 2026] unifies FinQA, ConvFinQA, and TAT-DQA into 23,088 queries over 7,318
73 financial documents; where the original paper tested six methods with two metrics, we evaluate nine
74 retrieval configurations together with a BGE-M3 robustness panel and a ColBERTv2 follow-up,
75 with a comprehensive set of retrieval and generation metrics.

76 **3 Methodology**

77 **3.1 Dataset**

78 T²-RAGBench [Strich et al., 2026] contains 23,088 question-context-answer triples from FinQA
79 (8,281), ConvFinQA (3,458), and TAT-DQA (11,349), covering 7,318 unique financial documents
80 averaging ~920 tokens each. Each document mixes text and markdown-formatted tables from SEC
81 filings. The benchmark reformulated all questions using Llama 3.3-70B to incorporate identifying

information (company name, sector, report year), producing questions with exactly one correct answer regardless of context. All answers are numerical. The original best method reached only 41% Number Match against an oracle ceiling of 72–79%.

3.2 Retrieval Methods

We evaluate nine retrieval configurations spanning five categories, together with three BGE-M3 configurations that vary the encoder and a ColBERTv2 follow-up. Full hyperparameters are in Appendix A; prompt templates in Appendix C.

BM25. Okapi BM25 [Robertson and Walker, 1994] with $k_1 = 1.2$, $b = 0.75$, providing strong lexical matching for domain-specific terminology.

Dense retrieval. We encode queries and documents with **OpenAI text-embedding-3-large** (3,072d) via Azure AI Foundry, indexing with FAISS IndexFlatIP for exact inner-product search.

Hybrid RRF. Fuses BM25 and dense ranked lists via Reciprocal Rank Fusion [Cormack et al., 2009] with $k = 60$.

Hybrid + Cohere Rerank. Two-stage pipeline: hybrid RRF retrieves 50 candidates, reranked by Cohere Rerank v4.0 Pro [Cohere, 2025], returning top 10.

HyDE. Generates a hypothetical answer passage via GPT-4.1-mini and retrieves with its embedding [Gao et al., 2023].

Multi-Query. Generates three query variants via GPT-4.1-mini, retrieves for each, and merges via RRF [Raudaschl, 2024].

Contextual Retrieval. Prepends LLM-generated context summaries to each document at indexing time [Anthropic, 2024]. We apply it to both the dense and hybrid pipelines, yielding two configurations (*Contextual Dense*, *Contextual Hybrid*) reported separately. With seven other methods listed above, this brings the count to nine retrieval configurations.

CRAG. Classifies retrieved documents as relevant/ambiguous/irrelevant; rewrites query and re-retrieves when confidence is low [Yan et al., 2024].

BGE-M3 robustness panel. To test whether the encoder choice changes the ranking, we evaluate three additional configurations built on the BGE-M3 dense encoder (1024-d) [Chen et al., 2024a] and the BAAI bge-reranker-v2-m3 cross-encoder: dense retrieval, hybrid RRF (BM25 + BGE-M3), and the corresponding two-stage Hybrid + BGE Rerank pipeline. Both rerankers run inside their fine-tune context windows; configuration details are in Section 3.3.

ColBERTv2 (late interaction). As a follow-up to the table-structure failure mode identified by our error taxonomy (Section 5), we additionally evaluate ColBERTv2 [Santhanam et al., 2022] as a single-method retriever. ColBERTv2 retains a per-token vector for every document position rather than pooling to a single embedding, so a table cell whose label and value sit in different cells of the same row can still match a query token-by-token. We use the official `colbert-ir/colbertv2.0` checkpoint via `ragatouille` with `max_document_length=512` (the value used during ColBERTv2 fine-tuning); documents longer than 512 tokens are split into passages, scored independently, and aggregated to document level by max passage score per query.

3.3 Evaluation

Retrieval: Recall@ k ($k \in \{1, 3, 5, 10, 20\}$), MRR@ k , nDCG@ k , MAP. **Generation:** Number Match (NM) with tolerance $\epsilon = 10^{-2}$. **Statistics:** Paired bootstrap tests ($B = 10,000$, Bonferroni-corrected, $p < 0.05$).

Category	Method	R@1	R@3	R@5	R@10	MRR@3	nDCG@10	MAP
Single-method	BM25 (sparse)	0.293	0.552	0.644	0.735	0.411	0.515	0.449
	Dense (text-embed-3-large)	0.248	0.481	0.587	0.703	0.351	0.466	0.398
	Dense (BGE-M3) [◊]	0.229	0.434	0.515	0.607	0.321	0.413	0.357
	ColBERTv2 (late interaction) [◊]	0.301	0.526	0.616	0.709	0.402	0.501	0.440
Query expansion	HyDE (gpt-4.1-mini)	0.221	0.441	0.544	0.671	0.318	0.433	0.365
	Multi-Query + RRF	0.283	0.539	0.640	0.734	0.397	0.506	0.439
Index augment.	Contextual Dense	0.266	0.508	0.615	0.732	0.373	0.490	0.420
	Contextual Hybrid	0.327	0.610	0.717	0.818	0.454	0.571	0.497
Adaptive	CRAG (gpt-4.1-mini)	0.302	0.556	0.658	0.788	0.415	0.536	0.456
Fusion + rerank	Hybrid (BM25+Dense, RRF)	0.308	0.588	0.695	0.801	0.433	0.551	0.477
	Hybrid (BGE-M3, RRF) [◊]	0.290	0.545	0.641	0.741	0.404	0.513	0.445
	Hybrid + Cohere Rerank	0.472	0.758	0.816	0.861	0.605	0.683	0.625
	Hybrid (BGE-M3) + BGE Rerank [◊]	0.383	0.674	0.751	0.822	0.516	0.612	0.547

Table 1: Main retrieval results on T²-RAGBench (23,088 queries, 7,318 documents). [◊]Open-weight rows run locally on a single GPU: BGE-M3 (BAAI/bge-m3, 1024-d) and BAAI/bge-reranker-v2-m3 for the BGE-M3 stack, and ColBERTv2 (colbert-ir/colbertv2.0, late interaction) as a separate single-method retriever. Unmarked rows use Azure AI Foundry endpoints (text-embedding-3-large, Cohere Rerank). Latency is omitted because cross-hardware comparison would be misleading. All within-stack pairwise differences are statistically significant at $p < 0.05$ after Bonferroni correction over the comparison family, with the exceptions noted in Section 4.2. Best in **bold**.

Setup. BM25 and FAISS run locally on Apple Silicon. Embedding, LLM calls, and Cohere reranking use Azure AI Foundry. The BGE-M3 robustness panel uses BAAI/bge-m3 (1024-d) [Chen et al., 2024a] and the BAAI/bge-reranker-v2-m3 cross-encoder, both running locally on a single NVIDIA RTX PRO 6000 Blackwell GPU (102 GB). Each model is configured at its officially recommended operating point: dense encoders at 8,192-token context (BGE-M3’s native maximum and approximately the OpenAI API limit), Cohere Rerank v4.0 Pro at max_tokens_per_doc=4,096 (default), and bge-reranker-v2-m3 at max_length=1,024 (BAAI’s stated fine-tuning length; the 512 used in the model-card example is a demo setting and truncates the 920-token mean T²-RAGBench document). Each candidate document is clipped to the first 4,096 characters before reranking. Candidate pools are 50 for Cohere and 60 for BGE, both above the 20-candidate threshold below which reranker quality degrades sharply (Section 4.4). Both rerankers run inside their trained context windows; we do not extend either beyond its fine-tune length. Latency is reported only within hardware-matched comparisons. All experiments use seed 42, temperature 0, and the standard train/test split. Code is publicly released [Anonymous Authors, 2026].

4 Results

4.1 Main Retrieval Results

Table 1 presents retrieval performance of all evaluated methods on the full T²-RAGBench test set (23,088 queries, 7,318 documents).

Hybrid RRF with a cross-encoder reranker is the clear winner: Recall@5 of 0.816, compared to 0.695 for unranked hybrid (+17%), 0.644 for BM25 (+27%), and 0.587 for dense (+39%). MRR@3 moves from 0.433 to 0.605 on the same transition, a 40% relative gain. Among the first-stage retrievers, BM25 beats dense on every metric except Recall@20, where they are effectively tied. Contextual retrieval adds 2–3 pp of Recall@5 to both dense and hybrid for an LLM call per document at indexing time. CRAG edges past BM25 (+1.4 pp Recall@5) but not past hybrid fusion. Per-subset numbers are in Appendix A.

Retrieval	Number Match
BM25	0.251
Dense	0.257
Hybrid RRF	0.282
Oracle	0.350

Table 2: End-to-end Number Match by retrieval method, GPT-4.1-mini generator, $\epsilon = 10^{-2}$ tolerance. Pearson $r = 0.980$ across these four points (Figure 6); BM25 and Dense are within 0.6 pp on NM, the only rank inversion versus Recall@5. The Oracle ceiling indicates retrieval headroom remains.

149 4.2 Negative Results: What Doesn’t Work and Why

150 **HyDE is counterproductive.** HyDE underperforms vanilla dense retrieval (Recall@5: 0.544 vs.
151 0.587, $p < 0.001$), reproducing the result in Strich et al. [2026]. The failure is specific to numerical
152 queries. Asked to draft a hypothetical answer for “What was the 2019 net revenue?”, the LLM
153 produces plausible-looking but fabricated figures; the resulting embedding is close to documents
154 that happen to mention those invented numbers, not to the one with the true answer. HyDE’s premise
155 is paraphrase matching, and exact numerical answers are the setting where paraphrase does the least
156 work.

157 **Multi-query provides negligible benefit.** Multi-query achieves Recall@5 of 0.640, statistically
158 indistinguishable from BM25’s 0.644 ($p = 0.31$). Financial queries are already specific; reformu-
159 lations do not surface new documents because the retrieval gap is structural (table format), not
160 lexical.

161 **CRAG cannot match hybrid fusion.** While 63% of queries triggered CRAG’s correction path-
162 way, rewritten queries face the same table structure challenge. Query rewriting alone cannot substi-
163 tute for combining complementary retrieval signals.

164 4.3 End-to-End Generation

165 Number Match tracks retrieval quality closely across the four retrievers (Table 2): BM25 and Dense
166 are within 0.6 pp NM (Dense 0.257 above BM25 0.251, the one rank inversion versus Recall@5 in
167 this panel), Hybrid RRF reaches 0.282, and the Oracle ceiling is 0.350. Pearson correlation against
168 Recall@5 across the four points is $r = 0.980$ (Figure 6); the inversion is at the noise floor and the
169 linear trend is otherwise tight. The Oracle–Hybrid gap of 6.8 pp is the room left after retrieval is
170 fixed; everything below is retrieval headroom. A linear extrapolation from the four-point fit projects
171 NM ≈ 0.31 at the Hybrid + Cohere Rerank operating point ($R@5 = 0.816$); we mark this as an
172 extrapolation rather than a measurement, since end-to-end NM was evaluated only at the four points
173 above.

174 4.4 Ablation Studies

175 **Fusion method.** Convex Combination with $\alpha = 0.5$ (equal weight on BM25 and dense) reaches
176 Recall@5 of 0.726, beating RRF ($k = 60$) at 0.695; within RRF, $k = 10$ is the best setting we tried
177 (0.716). Equal weighting wins in both cases (Figure 1).

178 **Reranker depth.** The dependence on candidate-pool size is a cliff, not a smooth ramp. At 20 can-
179 didates, hybrid RRF already places the gold document in the input pool 87.7% of the time ($R@20$,
180 Table 5), but the reranker recovers only 52% of those: $R@5 = 0.458$. At 50 candidates the rerank
181 $R@5$ jumps to 0.826, and at 100 it reaches 0.888 (Figure 2); the conversion rate rises sharply as
182 the pool fills out, with diminishing returns past 100.¹ We conjecture, without testing it directly,
183 that cross-encoders sharpen on candidate sets that span both clear positives and clear negatives, and

¹Hybrid $R@50$ and $R@100$ were not part of our metric sweep, so the exact conversion rates at the larger pool sizes are bounded above (by the same $R@N$ that they would denominate) but not directly measured. The 52% at pool=20 is exact: 0.458/0.877.

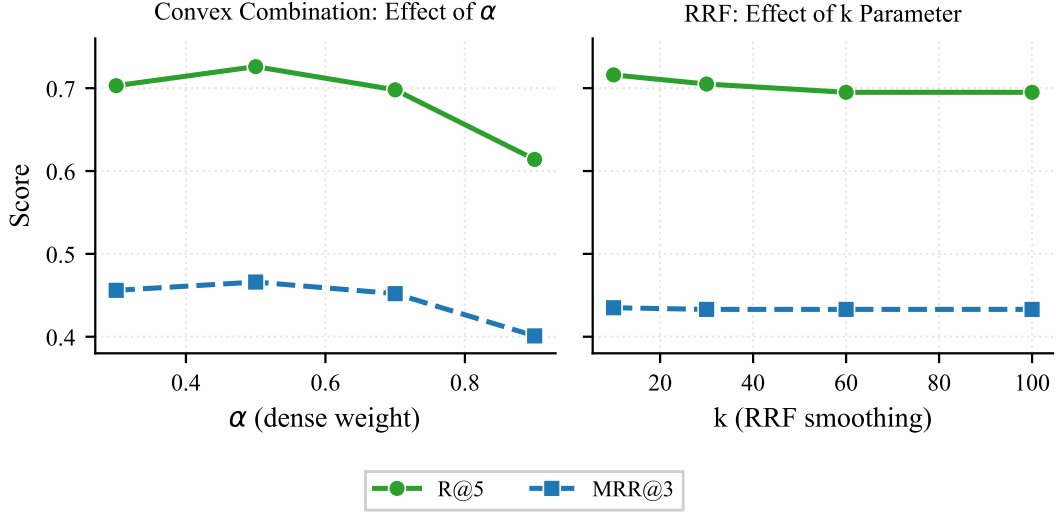


Figure 1: Fusion method ablation. Left: Convex Combination with varying α ; $\alpha = 0.5$ is optimal. Right: RRF with varying k ; lower k yields better results.

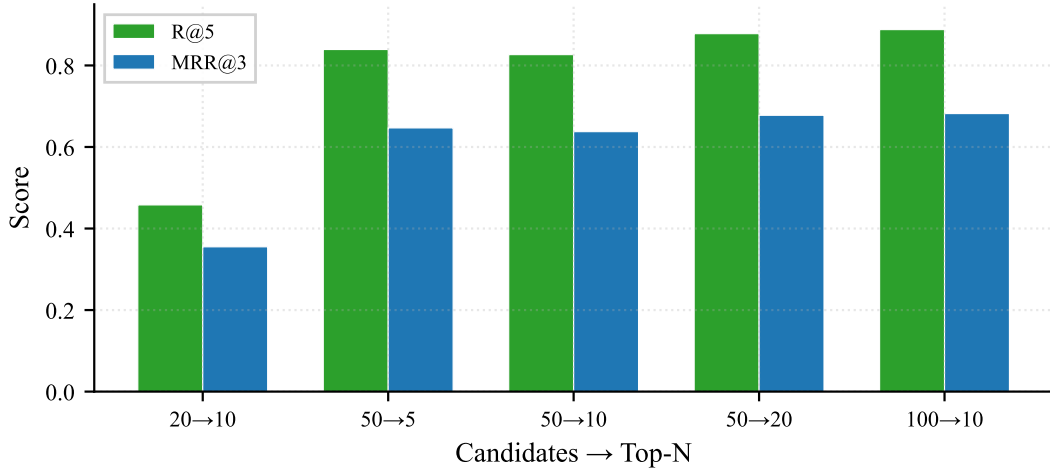


Figure 2: Reranker depth ablation. Critical threshold at 50 candidates.

184 that a short pool of all-relevant hybrid top-20 leaves the score margins between candidates noise-
 185 dominated. Practically, a candidate pool of at least 50 is the operating point we recommend.²

186 5 Error Taxonomy for Text-and-Table Retrieval

187 The 31.1% of queries where hybrid RRF misses the gold document in its top-5 is not a uniform error
 188 surface. We sampled 100 of those 7,188 failures and had GPT-5.4 label each by failure type; the
 189 resulting distribution (Table 3) is dominated by a single category.

190 **Table structure mismatch (73%).** Standard encoders see a table as linearized text; the two-
 191 dimensional row/column geometry is gone by the time the embedding is computed. A query like
 192 “What was net income in 2019?” cannot match a row whose label (“net income”) and fiscal year

²The 0.826 reported here is from the reranker-depth ablation; Table 1 reports 0.816 for the same configuration from an independent run of the same setup. Cohere Rerank is not bit-deterministic in our pipeline even at temperature 0, and the 1 pp gap between these two runs is the in-paper variance estimate; we treat it as the floor for any single-run comparison against the Cohere stack.

Failure Category	%
Table structure mismatch	73
Numerical reasoning	20
Vocabulary mismatch	5
Ambiguous query	1
Long document	1

Table 3: Error taxonomy ($n=100$ sampled failures). Table structure mismatch dominates; Wilson 95% CI for the 73% headline rate is [64%, 81%], and the category remains the modal failure across the entire interval.

193 (“2019”) sit in different cells of the same row. This is the open problem the benchmark surfaces:
194 table-aware embeddings or retrievers that preserve row/column structure.

195 **Numerical reasoning (20%)** failures occur when questions require computation (e.g., year-over-
196 year change) rather than direct lookup.

197 **Cross-subset patterns.** TAT-DQA has the highest failure rate (35.6% vs. 27.2% FinQA, 26.0%
198 ConvFinQA). Among failures, 71.0% of gold documents appear in *neither* the dense nor BM25
199 top-5, indicating genuinely hard cases rather than fusion artifacts.

200 **Implications for evaluation.** A method that pushes aggregate Recall@5 up might only be chip-
201 ping at the vocabulary mismatch tail (5% of errors) while leaving the table structure mass (73%)
202 untouched. Future evaluations on heterogeneous corpora should split metrics by failure category;
203 the aggregate alone is misleading.

204 6 Discussion

205 **Why BM25 wins, and what that means for leaderboards.** text-embedding-3-large ranks
206 among the top dense encoders on MTEB and BEIR, but on T²-RAGBench BM25 beats it on every
207 metric except Recall@20, where the two are tied. Financial queries are full of terms that appear
208 verbatim in answer documents (company names, tickers, standardized metric labels, fiscal period
209 identifiers), and lexical matching exploits this directly. A dense encoder distributes the same tokens
210 across a neighborhood of nearby vectors, many of which the query never asked for. The lesson
211 is narrower than “rethink evaluation”: MTEB and BEIR rankings do not transfer to corpora that
212 combine text and tables, and a dense encoder picked from those leaderboards is the wrong default
213 for such data.

214 **What changes when we swap to a fully open stack.** Swapping text-embedding-3-large for
215 the BGE-M3 dense encoder (1024-d) leaves the broad ranking intact but localises the gap. As a
216 single-method retriever, BGE-M3 is significantly weaker than both OpenAI dense ($R@5 = 0.515$
217 vs. 0.587, $p < 0.001$) and BM25 (0.515 vs. 0.644, $p < 0.001$). Hybrid fusion absorbs the encoder
218 difference: Hybrid (BM25 + BGE-M3, RRF) reaches $R@5 = 0.641$ against BM25 at 0.644, a paired-
219 bootstrap difference of -0.31 pp ($p = 0.265$, not significant). The same fusion with OpenAI dense
220 yields $+5.1$ pp over BM25 (0.695 vs. 0.644), so dense quality matters more for *amplifying* BM25
221 than for *matching* it. With both rerankers run at their fine-tune length, the two-stage results are
222 closer than the encoder gap suggests. Cohere Rerank lifts hybrid by $+12.1$ pp on the OpenAI stack
223 (0.695 $\rightarrow$ 0.816); bge-reranker-v2-m3 at `max_length=1,024` lifts hybrid by $+11.0$ pp on the BGE-
224 M3 stack (0.641 $\rightarrow$ 0.751, $p < 0.001$). The two rerankers move hybrid by nearly the same absolute
225 amount; what remains is a 6.5 pp gap (0.816 vs. 0.751) between the closed and open pipelines, well
226 below the per-subset variability we see in Appendix A. The 1,024 setting matters: at the model-card
227 example value of 512, the open reranker instead *reduces* hybrid $R@5$ by 3.1 pp (0.641 $\rightarrow$ 0.610,
228 $p < 0.001$). On a corpus with 920-token mean document length the apparent regression at 512 is
229 a truncation artifact, since BAAI’s fine-tuning recipe uses 1,024 tokens; the difference between the
230 two settings is a paired-bootstrap $+14.1$ pp ($p < 0.001$). We report the in-regime 1,024 number
231 throughout and treat the 512 baseline as a truncation diagnostic, not a quality claim about the model.

Late-interaction retrieval is a partial fix. The error taxonomy points at table-cell answers as the dominant failure mode and flags late-interaction retrievers as the natural representational fix. We tested ColBERTv2 [Santhanam et al., 2022] as a single-method retriever, with passages encoded at `max_doc_length=512` and aggregated to document-level by max passage score. ColBERTv2 lifts dense retrieval substantially ($R@5 = 0.616$ vs. 0.587 for `text-embedding-3-large`, $+2.9$ pp paired-bootstrap, $p < 0.001$; vs. 0.515 for BGE-M3, $+10.1$ pp, $p < 0.001$). It does not, however, beat BM25 (0.616 vs. 0.644 , -2.8 pp, $p < 0.001$) and sits well below hybrid fusion (-8.0 pp vs. Hybrid RRF, $p < 0.001$). Per-subset, ColBERTv2 ties BM25 on ConvFinQA (0.695 vs. 0.696) but loses on FinQA and TAT-DQA, and notably does *not* disproportionately help on TAT-DQA, the most table-heavy subset. Token-level late interaction is a partial fix: it closes much of the gap from a dense encoder to BM25 without reaching it, and the table-structure failure mode is not absorbed by replacing soft attention over a pooled vector with max-similarity over linearized tokens.

The reranker does most of the heavy lifting. Adding a cross-encoder reranker on top of hybrid RRF is worth $+17.2$ pp MRR@3 and $+12.1$ pp Recall@5. That is a larger improvement than anything else we measured, and it is the only configuration that closes most of the gap to the oracle ceiling. Candidate depth matters: the rerank step degrades below 50 candidates (Section 4.4).

If you have to pick one setup. Use hybrid RRF with a cross-encoder reranker and a candidate pool of at least 50. Contextual retrieval is worth the one-time indexing cost: it lifts both dense and hybrid by roughly 2–3 pp of Recall@5 for an LLM call per document. Skip HyDE and multi-query expansion for numerical domains: HyDE actively moves the numbers in the wrong direction, and multi-query does not close the gap to BM25. Do not extrapolate from MTEB or BEIR; evaluate on data that looks like yours.

7 Conclusion

On a financial QA benchmark of 23,088 queries that mix prose and tables, hybrid RRF followed by a cross-encoder reranker delivers the best end-to-end performance; stacking query expansion on top of a dense encoder makes things worse. Two findings are worth carrying to other heterogeneous corpora. BM25 beats a state of the art dense encoder on nearly every metric, so MTEB and BEIR rankings should not be taken as guidance for data this isn’t drawn from. And 73% of the retrieval failures that remain share a single cause: the answer is a table cell, and the encoder processes tables as serialized text. Closing that gap is less about scaling the encoder than about representing tabular structure. Token-level late interaction (ColBERTv2) is a partial fix — it lifts dense retrieval substantially but does not beat BM25 and does not solve the table-cell case. Promising directions we did not explore are structure-aware chunking, table-aware encoders such as Jina-ColBERT-v2 [Sturua et al., 2024] or TAPAS-style architectures, and using late-interaction inside hybrid fusion or as a reranker rather than as a single-method retriever.

Limitations

One domain. The corpus is financial filings. A different domain (legal briefs with tables, medical records, scientific papers) may shift both the absolute numbers and the ordering between methods, though we expect BM25’s edge to be strongest where terminology is standardized and weakest where paraphrase is common.

Numerical answers only. Every answer in T²-RAGBench is a number, which is why Number Match is the primary generation metric. We do not test whether the same retrieval ordering holds for free-form answers. End-to-end NM was measured at four operating points (Dense, BM25, Hybrid RRF, Oracle); the two-stage pipelines (Hybrid + Cohere Rerank, Hybrid (BGE-M3) + BGE Rerank) and the BGE-M3 single-method panel were evaluated on retrieval only. The linear projection in Section 4.3 is an extrapolation, not a measurement.

Whole-document retrieval. We index each document whole (~ 920 tokens). Chunking changes the recall game; it may close part of the dense encoder gap on longer documents by giving the encoder a smaller surface to summarize.

Two dense encoders, two rerankers. We evaluated two dense encoders (text-embedding-3-large, BGE-M3) and two cross-encoder rerankers (Cohere Rerank v4.0 Pro, bge-reranker-v2-m3). The qualitative ordering (BM25 above either dense encoder, hybrid above both, hybrid+rerank above hybrid) held in every pairing we tested once both rerankers ran inside their fine-tune context windows (4,096 and 1,024 tokens respectively). Encoder families with explicit table-aware pretraining may behave differently, and we do not claim universality beyond either reranker’s trained regime.

Commercial APIs. The dense encoder, reranker, and generation LLM are all commercial endpoints. Model weights and behaviors can change. We release per-query outputs so that our results can be compared against future runs rather than re-reproduced from scratch.

Ethics Statement

The source data is public: SEC filings, accessed through the T²-RAGBench release. No personal, private, or human-subject data is used, and all LLM calls go through commercial endpoints under their standard terms. The main positive externality we expect is cheaper, more accurate retrieval in RAG systems built over regulatory or financial documents. The main negative externality to flag is that better retrieval over SEC filings could make it easier to automate certain information asymmetry exploits (e.g., faster-than-human extraction of earnings figures for algorithmic trading). The mitigation is identical to the existing state of the art: these documents are already public, and practitioners can already search them. Releasing a stronger retrieval baseline does not change who has access; it changes only how fast the useful signal can be surfaced.

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Justification: Section 3.3 describes the experimental setup. Appendix A provides complete hyperparameter tables, including the operating-point values that govern fair comparison across encoders and rerankers (`max_seq_length`, `max_tokens_per_doc`, `max_length`, candidate pool size). Appendix C provides all prompt templates. Each released per-query result JSON also embeds a provenance block recording the git SHA, library versions (`torch`, `transformers`, `sentence-transformers`, `faiss`), CUDA version, GPU name, HuggingFace model commit SHA, and the SHA-256 of the FAISS index used. Random seed, temperature, and all configuration details are specified.

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Justification: The benchmark code and evaluation scripts are released as an open-source repository. The dataset (T²-RAGBench) is publicly available from the original authors.

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480 per include the full text of instructions given to participants and screenshots, if applicable,
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497 Answer: [Yes]

498 Justification: LLMs are core components of several retrieval methods (HyDE, Multi-Query,
499 CRAG, Contextual Retrieval) and the generation evaluation stage. All LLM usage is doc-
500 umented in Sections 3.2–4.3 with model names, temperatures, and exact prompts in Ap-
501 pendix C.

502 A Hyperparameter Details and Full Results

503 Table 4 lists all hyperparameters. Table 5 presents full retrieval metrics. Table 6 shows per-subset
504 performance.

505 B Additional Figures

506 C Prompt Templates

507 All prompts use gpt-4.1-mini via Azure AI Foundry.

508 C.1 Generation Prompt

509 Answer the following question based ONLY on
510 the provided context.

511 If the answer is a number, provide just the
512 number. If you cannot answer from the
513 context, say "UNANSWERABLE".

514

515 Context: {context}

516 Question: {question}

517 Answer:

518 C.2 HyDE Prompt

519 Given the following question about financial
520 data, write a short passage that would

Method	Parameter	Value	Notes
BM25	k_1 / b Tokenizer	1.2 / 0.75 whitespace split	Canonical Okapi defaults Via <code>rank_bm25</code>
Dense (text-embed-3-large)	Dimensions Max input Index type	3,072 ~8,191 tokens FAISS IndexFlatIP	OpenAI via Azure AI Foundry API default Exact cosine search
Dense (BGE-M3) [◇]	Dimensions <code>max_seq_length</code> Index type	1,024 8,192 FAISS IndexFlatIP	BAAI/bge-m3 via <code>sentence-transformers</code> BGE-M3 native maximum Exact cosine search
Hybrid RRF	k	60 (ablation: 10, 30, 100)	Smoothing constant
Hybrid CC	α	0.5 (ablation: 0.3, 0.7, 0.9)	Dense weight
Cohere Rerank	Model / top_n <code>max_tokens_per_doc</code> Candidate pool	v4.0-pro / 10 4,096 50	Via Azure AI Foundry API default From first stage
BGE Rerank [◇]	Model / top_n <code>max_length</code> Candidate pool	<code>bge-reranker-v2-m3</code> / 10 1,024 60	Local cross-encoder BAAI fine-tune length From first stage; minor asymmetry vs Cohere (50)
HyDE	LLM / temp / max_tok	<code>gpt-4.1-mini</code> / 0 / 150	Single generation
Multi-Query	Variants / RRF k	3 / 60	<code>gpt-4.1-mini</code> , temp 0
CRAG	Eval temp / rewrite temp	0.0 / 0.5	<code>gpt-4.1-mini</code>
Contextual	LLM / temp / max_tok	<code>gpt-4.1-mini</code> / 0 / 100	Per-document context
Global	Seed / top- k Bootstrap	42 / {1,3,5,10,20} $B=10,000, p<0.05$	All methods Bonferroni-corrected

Table 4: Complete hyperparameter settings for all retrieval methods.

Method	R@1	R@3	R@5	R@10	R@20	MRR@3	nDCG@10	MAP
Dense (BGE-M3) [◇]	0.229	0.434	0.515	0.607	0.688	0.321	0.413	0.357
HyDE	0.221	0.441	0.544	0.671	0.767	0.318	0.433	0.365
Dense	0.248	0.481	0.587	0.703	0.798	0.351	0.466	0.398
Contextual Dense	0.266	0.508	0.615	0.732	0.817	0.373	0.490	0.420
ColBERTv2 (late interaction) [◇]	0.301	0.526	0.616	0.709	0.780	0.402	0.501	0.440
Multi-Query	0.283	0.539	0.640	0.734	0.820	0.397	0.506	0.439
Hybrid (BGE-M3) [◇]	0.290	0.545	0.641	0.741	0.819	0.404	0.513	0.445
BM25	0.293	0.552	0.644	0.735	0.797	0.411	0.515	0.449
CRAG	0.302	0.556	0.658	0.788	0.788 [†]	0.415	0.536	0.456
Hybrid RRF	0.308	0.588	0.695	0.801	0.877	0.433	0.551	0.477
Contextual Hybrid	0.327	0.610	0.717	0.818	0.887	0.454	0.571	0.497
Hybrid (BGE-M3) + BGE Rerank [◇]	0.383	0.674	0.751	0.822	0.867	0.516	0.612	0.547
Hybrid + Rerank	0.472	0.758	0.816	0.861	*	0.605	0.683	0.625

Table 5: Full retrieval results sorted by nDCG@10. Rows marked [◇] use open-weight retrievers run locally on a single GPU (BGE-M3 stack or ColBERTv2); unmarked rows use text-embedding-3-large via Azure. *Rerank returns ≤ 10 documents. [†]CRAG returns ≤ 10 documents; R@20 set equal to R@10 for completeness.

521 contain the answer. Include specific numbers
522 and financial terms.
523 Question: {query}
524 Passage:

525 C.3 Multi-Query Prompt

526 You are a helpful assistant that generates
527 alternative search queries. Given the
528 following question, generate {n} alternative
529 phrasings that capture the same information
530 need but use different wording or
531 perspectives. Return each query on its own
532 line, numbered. Do not include any other text.

Subset	Method	R@5	R@10	MRR@3
FinQA	BM25	0.729	0.834	0.389
	Dense	0.611	0.748	0.308
	Hybrid	0.737	0.856	0.389
	Dense (BGE-M3) [◊]	0.611	0.715	0.373
	Hybrid (BGE-M3) + BGE Rerank [◊]	0.829	0.891	0.529
	Hybrid + Cohere Rerank	0.860	0.909	0.578
ConvFinQA	BM25	0.696	0.781	0.500
	Dense	0.654	0.781	0.410
	Hybrid	0.754	0.850	0.519
	Dense (BGE-M3) [◊]	0.585	0.689	0.332
	Hybrid (BGE-M3) + BGE Rerank [◊]	0.826	0.882	0.600
	Hybrid + Cohere Rerank	0.874	0.907	0.671
TAT-DQA	BM25	0.566	0.649	0.400
	Dense	0.549	0.647	0.364
	Hybrid	0.647	0.746	0.438
	Dense (BGE-M3) [◊]	0.423	0.503	0.279
	Hybrid (BGE-M3) + BGE Rerank [◊]	0.672	0.753	0.481
	Hybrid + Cohere Rerank	0.766	0.812	0.605

Table 6: Per-subset results. Hybrid + Cohere Rerank is the per-subset winner on every metric we report; among the two-stage open pipelines, BGE-M3 + BGE Rerank is consistently second and lifts the BGE-M3 hybrid base by 11–19 pp R@5 per subset. TAT-DQA is the hardest subset across the board. Rows marked [◊] use the BGE-M3 stack (BAAI/bge-m3 + bge-reranker-v2-m3).

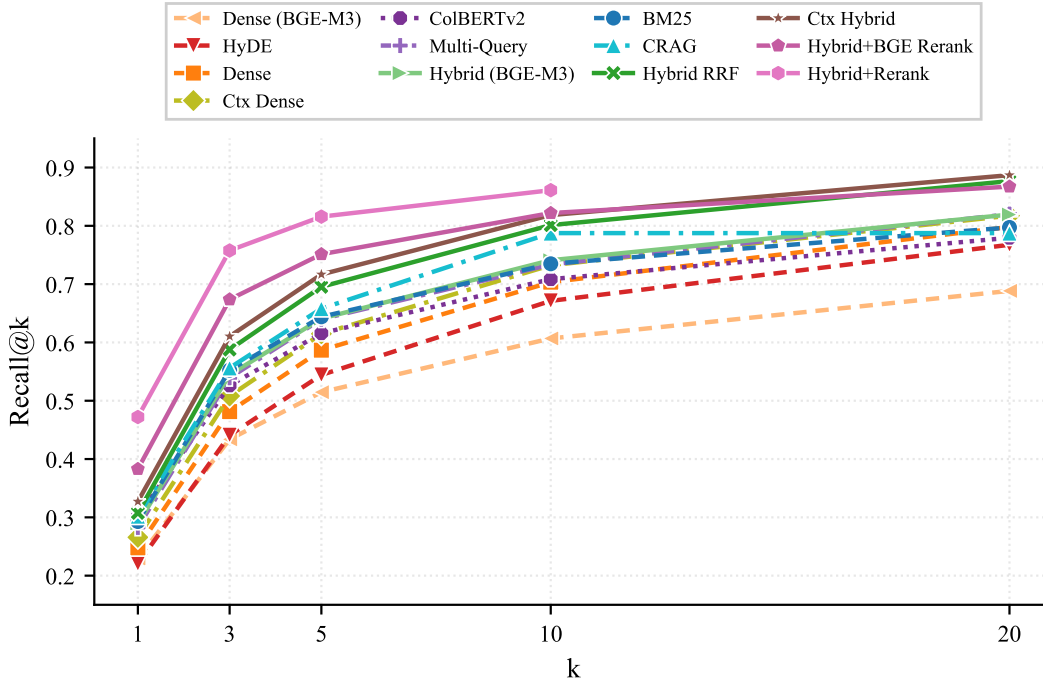


Figure 3: Recall@ k curves. Hybrid fusion outperforms both baselines at every k , with the largest gains at small k .

533
534 Original question: {query}
535 Alternative queries:

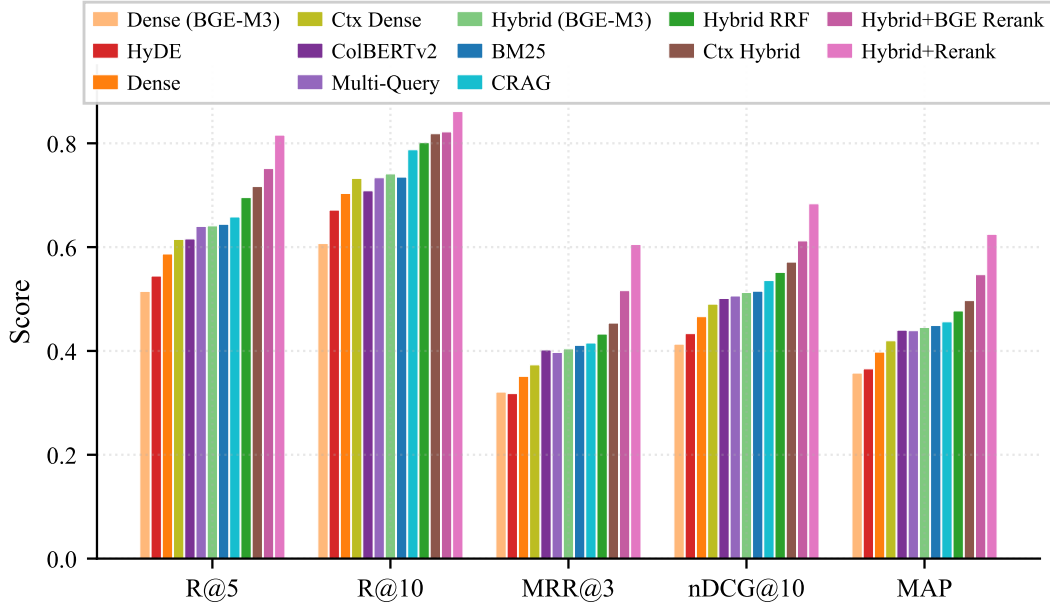


Figure 4: Grouped comparison across five metrics. BM25 outperforms dense retrieval; hybrid RRF dominates.

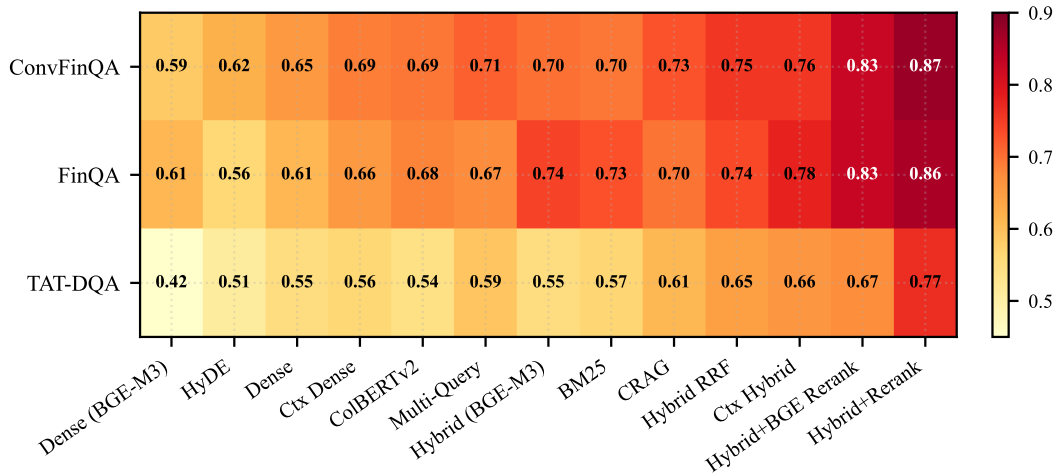


Figure 5: Recall@5 heatmap across methods and subsets. TAT-DQA is consistently hardest.

536 C.4 CRAG Evaluation Prompt

537 You are a relevance evaluator. Given a
 538 question and a retrieved document, classify
 539 the document's relevance.

540
 541 Question: {query}
 542 Document: {document}

543
 544 Respond with exactly one of:

- 545 - RELEVANT
- 546 - AMBIGUOUS
- 547 - IRRELEVANT

548
 549 Classification:

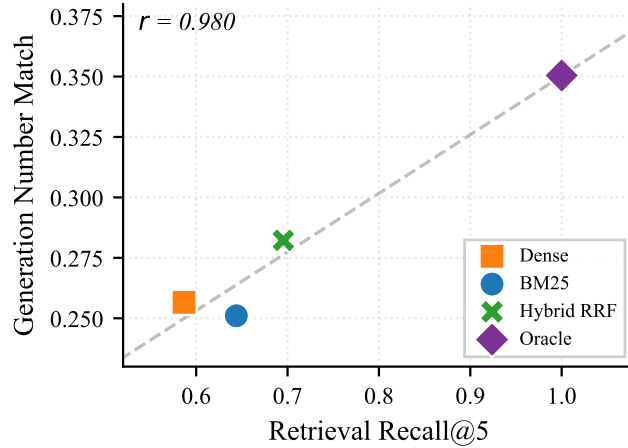


Figure 6: Retrieval and generation correlation ($r = 0.980$, $n = 4$): better retrieval tracks better answers. Four-point ablation (Dense / BM25 / Hybrid RRF / Oracle) on the GPT-4.1-mini generator. We did not measure NM end-to-end for Hybrid + Cohere Rerank or for the BGE-M3 stacks; the linear projection from this fit is reported in Section 4.3.

550 C.5 CRAG Rewrite Prompt

551 The following question was used to search a
 552 financial document corpus, but the retrieved
 553 results were not sufficiently relevant.

554
 555 Original question: {query}

556
 557 Please rewrite this question to be more
 558 specific and likely to retrieve the correct
 559 financial document.

560
 561 Rewritten question:

562 C.6 Contextual Retrieval Prompt

563 Here is a document:

564 <document>

565 {document}

566 </document>

567

568 Please provide a concise summary context
 569 (2-3 sentences) that captures the key
 570 topics and entities in this document, for
 571 the purpose of improving search retrieval.
 572 Answer only with the context, nothing else.